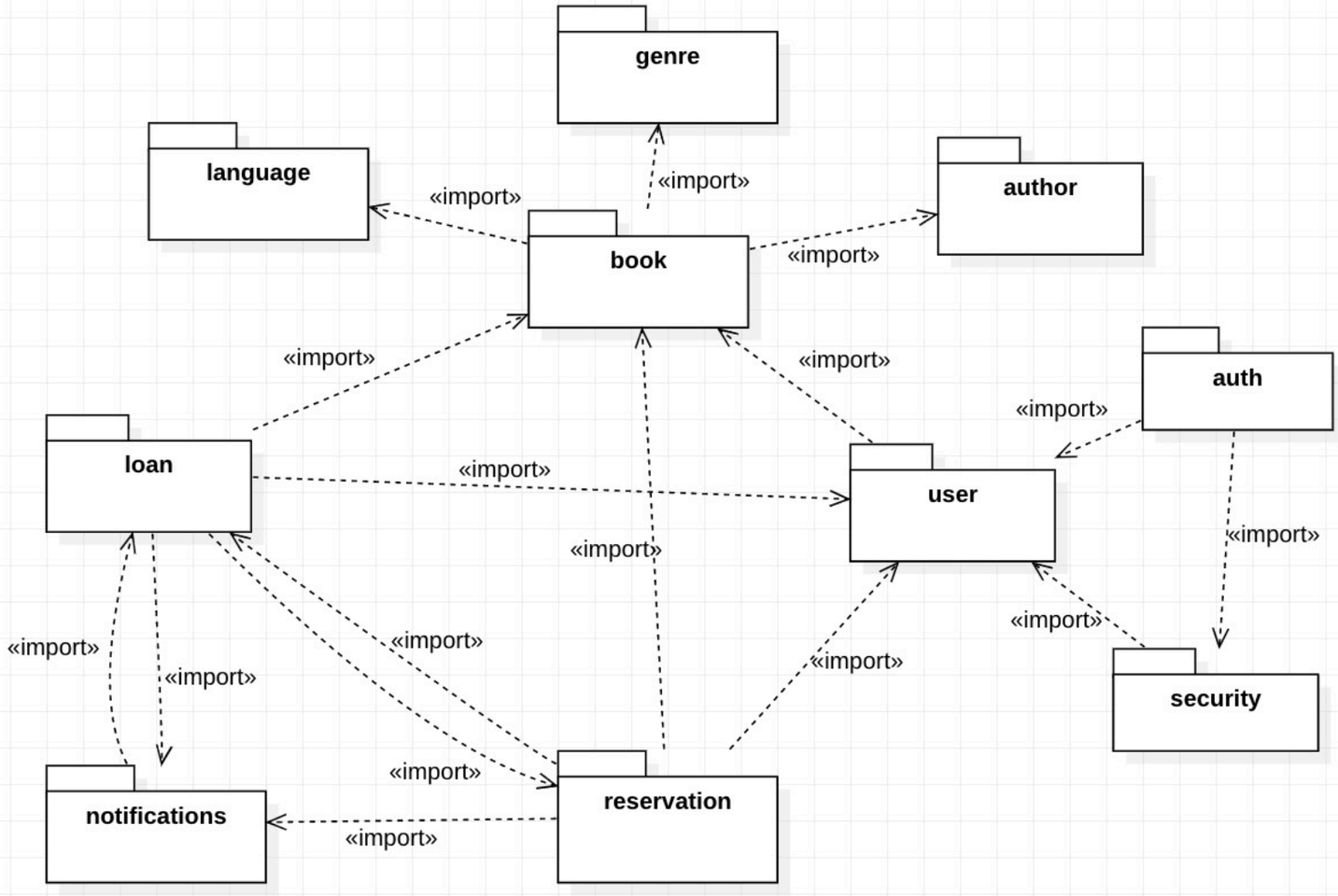

Text description:
This diagram shows feature-based modularization where packages import other packages they depend on. Repository interfaces (UserRepository, BookRepository, AuthorRepository, LoanRepository, ReservationRepository, GenreRepository, LanguageRepository) were chosen as the primary interface layer. These interfaces extend Spring Data JPA's CrudRepository and define database operations for their respective entities. This approach allows the database implementation to be replaced without modifying the service layer.

Packages:
user – Contains all classes related to user management, including the User entity, UserController, UserService, UserRepository, and related DTOs.
book – Contains classes for managing library book catalog, including the Book entity, BookController, BookService, BookRepository and book related DTOs. Handles book browsing searching and filtering.
author – Contains the Author entity, AuthorController, AuthorRepository and AuthorDTO. Manages book authors and their relationship to book through the many-to-many "writes" association.
genre – Contains the Genre entity, GenreController, GenreRepository and GenreDTO. Manages book genres used for filtering and categorization.
language – Contains Language entity, LanguageController, LanguageRepository and LanguageDTO. Manages book language used for filtering.
loan - Contains classes for handling book loans, including the Loan entity, LoanController, LoanService, LoanRepository, and loan-related DTOs. Manages loan creation, return and automatic due date handling.
reservation - Contains classes for handling book reservations, including the Reservation entity, ReservationController, ReservationService, ReservationRepository, and reservation related DTOs.
notifications – Contains NotificationService and MailService, responsible for sending email notifications to users about due dates, available reserved books.
auth – Contains authentication related classes including AuthController, AuthService and related DTOs. Handles user registration and login.
security – Contains security infrastructure classes including JwtUtil, SecurityConfig, JwtAuthenticationFilter and AuthorizationUtil. Provides JWT-based authentication and authorization across the application.